

Signals

1.

```
harshika@siri: ~/OSLab/Signa x + v
harshika@siri:~$ mkdir OSLab
mkdir: OSLab: File exists
harshika@siri:~$ cd OSLab
harshika@siri:~/OSLab$ mkdir Signals
harshika@siri:~/OSLab$ cd Signals
harshika@siri:~/OSLab/Signals$ gedit signal1.c

** (gedit:752): WARNING **: 06:47:58.346: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:752): WARNING **: 06:47:58.346: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~/OSLab/Signals$ gcc signal1.c -o signal1
harshika@siri:~/OSLab/Signals$ ./signal1
Program is running...
Program is running...
Program is running...
Program is running...
^C
SIGINT received!
```

```
1#include <stdio.h>
2#include <signal.h>
3#include <unistd.h>
4void handle_signal(int sig)
5{
6printf("\nSIGINT received!\n");
7}
8int main()
9{
10signal(SIGINT, handle_signal);
11while (1)
12{
13printf("Program is running...\n");
14sleep(2);
15}
16return 0;
17}
```

2.

```
harshika@siri:~/OSLab/Signa x + v
harshika@siri:~/OSLab/Signals$ gedit signal2.c
c

** (gedit:1440): WARNING **: 06:50:49.210: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:1440): WARNING **: 06:50:49.211: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~/OSLab/Signals$ gcc signal2.c -o signal2
harshika@siri:~/OSLab/Signals$ ./signal2
Program is running...
Program is running...
Program is running...
^C
```

```
1#include <stdio.h>
2#include <unistd.h>
3int main()
4{
5while (1)
6{
7printf("Program is running...\n");
8sleep(2);
9}
10return 0;
11}
```

3.

```
harshika@siri:~/OSLab/Signa x + v
harshika@siri:~/OSLab/Signals$ gedit signal3.c
c

** (gedit:1898): WARNING **: 06:52:37.406: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:1898): WARNING **: 06:52:37.407: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~/OSLab/Signals$ gcc signal3.c -o signal3
harshika@siri:~/OSLab/Signals$ ./signal3
Parent is waiting...
Child is running...
Child is terminating...
SIGCHLD received: Child has terminated.
Parent finished.
harshika@siri:~/OSLab/Signals$ gedit signal3.c

** (gedit:2094): WARNING **: 06:53:26.292: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:2094): WARNING **: 06:53:26.292: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
```

```
1#include <stdio.h>
2#include <signal.h>
3#include <unistd.h>
4#include <sys/wait.h>
5void handle_sigchld(int sig)
6{
7printf("SIGCHLD received: Child has terminated.\n");
8}
9int main()
10{
11signal(SIGCHLD, handle_sigchld);
12pid_t pid = fork();
13if (pid == 0)
14{
15// Child
16printf("Child is running...\n");
17sleep(3);
18printf("Child is terminating...\n");
19}
20else
21{
22// Parent
23printf("Parent is waiting...\n");
24wait(NULL);
25printf("Parent finished.\n");
26}
27return 0;
28}
```

4.

```
harshika@siri:~/OSLab/Signa x + v
** (gedit:2094): WARNING **: 06:53:26.292: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~/OSLab/Signals$ gedit signal4.c

** (gedit:2392): WARNING **: 06:54:32.510: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:2392): WARNING **: 06:54:32.511: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~/OSLab/Signals$ gcc signal4.c -o signal4
harshika@siri:~/OSLab/Signals$ ./signal4
Writing to pipe...
```

```
1#include <stdio.h>
2#include <unistd.h>
3int main()
4{
5int fd[2];
6pipe(fd);
7close(fd[0]); // Close read end
8printf("Writing to pipe...\n");
9write(fd[1], "Hello", 5);
10printf("Write completed.\n");
11return 0;
12}
```

5.

```
harshika@siri: ~/OSLab/Signa x + v
: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~/OSLab/Signals$ gedit signal5.c
c

** (gedit:2844): WARNING **: 06:56:49.771: Could not load Peas repository: T
ypelib file for namespace 'Peas', version '1.0' not found

** (gedit:2844): WARNING **: 06:56:49.771: Could not load PeasGtk repository
: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~/OSLab/Signals$ gcc signal5.c
-o signal5
harshika@siri:~/OSLab/Signals$ ./signal5
Child waiting...
Parent sending signal...
Child received SIGUSR1
harshika@siri:~/OSLab/Signals$ gedit signal5.c

** (gedit:3036): WARNING **: 06:57:29.790: Could not load Peas repository: T
ypelib file for namespace 'Peas', version '1.0' not found

** (gedit:3036): WARNING **: 06:57:29.790: Could not load PeasGtk repository
: Typelib file for namespace 'PeasGtk', version '1.0' not found

1#include <stdio.h>
2#include <signal.h>
3#include <unistd.h>
4#include <sys/wait.h>
5void handler(int sig)
6{
7printf("Child received SIGUSR1\n");
8}
9int main()
10{
11pid_t pid = fork();
12if (pid == 0)
13{
14// Child
15signal(SIGUSR1, handler);
16printf("Child waiting...\n");
17while (1)
18{
19sleep(1);
20}
21}
22else
23{
24// Parent
25sleep(2);
26printf("Parent sending signal...\n");
27kill(pid, SIGUSR1);
28sleep(2);
29kill(pid, SIGTERM);
30wait(NULL);
31}
32return 0;
33}
```

6.

```
harshika@siri: ~/OSLab/Signa x + v
: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~/OSLab/Signals$ gedit signal6.c
c

** (gedit:3361): WARNING **: 06:58:14.394: Could not load Peas repository: T
ypelib file for namespace 'Peas', version '1.0' not found

** (gedit:3361): WARNING **: 06:58:14.394: Could not load PeasGtk repository
: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~/OSLab/Signals$ gcc signal6.c
-o signal6
harshika@siri:~/OSLab/Signals$ ./signal6
Program started
Aborted
./signal6

1#include <stdio.h>
2#include <stdlib.h>
3int main()
4{
5printf("Program started\n");
6abort();
7printf("Program ended\n");
8return 0;
9}
```

7.

```
harshika@siri: ~/OSLab/Signa x + v
: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~/OSLab/Signals$ gedit signal7.c
c

** (gedit:3857): WARNING **: 06:59:30.801: Could not load Peas repository: T
ypelib file for namespace 'Peas', version '1.0' not found

** (gedit:3857): WARNING **: 06:59:30.802: Could not load PeasGtk repository
: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~/OSLab/Signals$ gcc signal7.c
-o signal7
harshika@siri:~/OSLab/Signals$ ./signal7
Before error
Segmentation fault
./signal7
harshika@siri:~/OSLab/Signals$ gedit signal7.c

** (gedit:4054): WARNING **: 07:00:17.899: Could not load Peas repository: T
ypelib file for namespace 'Peas', version '1.0' not found

** (gedit:4054): WARNING **: 07:00:17.900: Could not load PeasGtk repository
: Typelib file for namespace 'PeasGtk', version '1.0' not found

1#include <stdio.h>
2int main()
3{
4int *p = NULL;
5printf("Before error\n");
6*p = 10;
7printf("After error\n");
8return 0;
9}
10
```